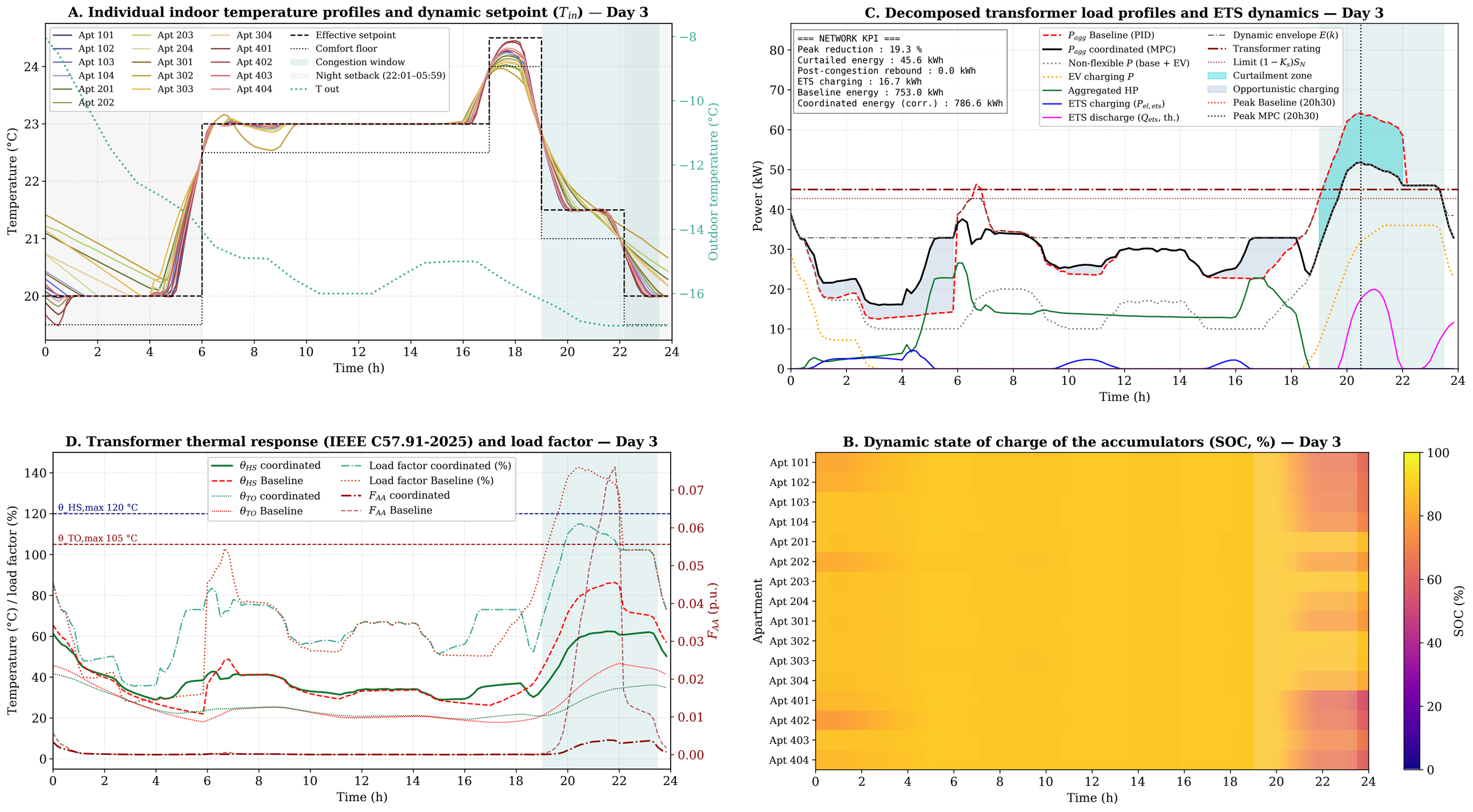

A. Individual indoor temperature profiles and dynamic setpoint (T_{in}) — Day 3
Apt 101
Apt 102
Apt 103
Apt 104
Apt 201
Apt 202
Apt 203
Apt 204
Apt 301
Apt 302
Apt 303
Apt 304
Apt 401
Apt 402
Apt 403
Apt 404
Effective setpoint
Comfort floor
Congestion window
Night setback (22:01–05:59)
T out
Temperature (°C)
Outdoor temperature (°C)
Time (h)
C. Decomposed transformer load profiles and ETS dynamics — Day 3
=== NETWORK KPI ===
Peak reduction : 19.3 %
Curtailed energy : 45.6 kWh
Post-congestion rebound : 0.0 kWh
ETS charging : 16.7 kWh
Baseline energy : 753.0 kWh
Coordinated energy (corr.) : 786.6 kWh
P_{agg} Baseline (PID)
P_{agg} coordinated (MPC)
Non-flexible P (base + EV)
EV charging P
Aggregated HP
ETS charging ($P_{el,ets}$)
ETS discharge (Q_{ets}, th.)
Dynamic envelope $E(k)$
Transformer rating
Limit $(1-K_s)S_N$
Curtailment zone
Opportunistic charging
Peak Baseline (20h30)
Peak MPC (20h30)
Power (kW)
Time (h)
D. Transformer thermal response (IEEE C57.91-2025) and load factor — Day 3
θ_{HS} coordinated
θ_{HS} Baseline
θ_{TO} coordinated
θ_{TO} Baseline
Load factor coordinated (%)
Load factor Baseline (%)
F_{AA} coordinated
F_{AA} Baseline
θ_HS,max 120 °C
θ_TO,max 105 °C
Temperature (°C) / load factor (%)
F_{AA} (p.u.)
Time (h)
B. Dynamic state of charge of the accumulators (SOC, %) — Day 3
Apartment
SOC (%)
Time (h)